

Cross-Domain Detection of AI-Generated Songs: Benchmarking Encoder Models on Bahasa Indonesia

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Abstract—In recent years, advances in deep learning have significantly improved our ability to detect highly realistic AI-generated song content. This progress has raised concerns about copyright protection, cultural authenticity, and the spread of misleading content. Despite growing global attention, there are still very few research studies related to this domain, especially in low-resource languages such as Bahasa Indonesia, due to the lack of appropriate datasets. To address this gap, we compiled and gathered a dataset of Indonesian songs collected from publicly available sources on social media, consisting of both real performances and AI-generated modifications, reflecting real situations in which AI-generated content is usually shared. We investigate our detection strategies by comparing several detection techniques using established audio encoders such as AST, Wave2Vec2, and WavLM. These models are then evaluated on unseen and different domain datasets. This is a crucial challenge given the rapid growth of AI music generation technologies. This experiment highlights both techniques and offers new insights into cross-generator detection, as well as dataset design and model robustness, which are important for protecting musical integrity in the era of generative AI.

I. INTRODUCTION

The fast development of AI has given rise to generative AI technology, which has improved the results of AI-generated songs to be very similar to synthetic audio, images, and video [1]. These deepfakes use Generative Adversarial Networks (GANs), Transformer-based models, and Reinforcement Learning to generate results that are very similar to what humans create. This content can include actions, voices, and facial expressions. In the music domain, systems like Suno AI¹ show that current deep learning methods are able to mimic human creativity by analyzing large amounts of musical data and recreating compositions that are stylistically diverse and rich [2]. Although these technologies have great potential in the entertainment, learning, and creative sectors, they can also be misused by people who exploit the system in negative ways, such as digital identity theft, copyright violations, and cultural authenticity crises. [3], [4].

As deepfake technology has become more precise and easily accessible, its misuse has grown significantly. In Indonesia especially, adversaries are now able to produce synthetic songs in both Bahasa Indonesia that closely mimic human performances. This can be used for fraudulent purposes, to spread disinformation, or to impersonate artists [5], [6]. Manipulated songs or audio recordings of celebrities can be easily spread on social media to influence public opinion, disrupt cultural authenticity, and be politically misused. In the music industry, AI-generated songs also present a great challenge in terms of copyright and ethical concerns around originality and ownership [7].

Current AI-generated detection systems have the ability to detect, and most of them have been trained and tested exclusively on English-language datasets and mainly focus on speech rather than music. This represents a significant limitation, especially in linguistically diverse countries like Indonesia, where detection systems are unlikely to generalize across different dialects, accents, and cultural contexts. We address this gap by focusing on AI-generated songs. We gather and compile a dataset consisting of real and AI-generated music collected from publicly available social media platforms, reflecting strong real-case scenarios where such content is typically shared.

This research aims to solve the emerging threat of AI-generated songs that focus on Bahasa Indonesia, the objective of which is to study which detection model can efficiently capture and distinguish between authentic and artificially generated music content. Using Existing encoder and signal processing method [8], we implement and fine-tuned [9] such a detection system that could support digital forensics, help media companies in authenticating music content shared online, and help streaming platforms and authorities to safeguard copyright royalty which is currently being closely monitored in Indonesia [10] [11]. Furthermore, to assess cross-domain generalization we extended our evaluation test on Bahasa Indonesia cloned-speech dataset. This will enable us to understand how well these encoders models transfer from

¹Suno AI: www.suno.com

musical audio to spoken audio within the same linguistic domain.

II. LITERATURE REVIEW

A. How deepfake audio is Generated?

AI-generated music is produced through sophisticated machine learning techniques designed to generate and replicate the creativity and features of human compositions. Some of the methods that originally were developed for speech have been also implemented for music, since both of them rely on the same sequential modeling of acoustic signals. Two of the most popular are symbolic generation and audio-based generation. These methods create music using different representation such as MIDI, piano rolls, or scores, while direct synthesis waveforms was used to create speech audio which similar to human generation [12].

Several methods were build on complex and deep architecture, Such as Recurrent Neural Networks(RNNs), Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs) [13], and more recently Transformer-based architectures [14]. These models have been widely used for melody and music generation. GANs also show effectiveness in producing melodies and harmonies that highly correlated to it's training data. Lately, transformers-based encoder-decoders excel at learning long term dependencies across entire compositions, it's attention mechanism allows it to learn features from the entire audio representations, like how they model sequences in natural language.

Several state of the art models illustrate the evolution of AI-generated music. Early models such as those basic algorithms lacked of complexity and capabilities. With the current existence of model, Suno AI has combined Transformers, GANs, and Reinforcement Learning into a unified promising framework . This enables the framework to generate statistically divers audio generation and mimic human emotions. This proves that current technology appears to be very good and hard for human to differentiate between synthetic and human generated songs.

With the rise of commercial tools and open platforms, AI-generated music is highlight accessible at internet. The extensive research at AI and deep learning models also contribute to the current exponential of growth lowering the barrier of entry for developers to develop their own system. These publically accessed system allow everyone with the access of internet to generate music, jingles, or full length songs across multiple genres, increasing professional-quality music production. This brings the need to developed strong and reliable detection systems, as distinguish between human-composed and AI generated music is becoming increasingly difficult.

B. Current Detection Methods

Detection of AI-Generated audio is a challenging task since it requires identifying very subtle differences between synthetic and real speech. Researchers have been working on this problem over the past few years and have proposed a wide range of methods, from handcrafted feature analysis

to sophisticated deep learning-based methods to tackle this problem. The detection techniques typically fall into two broad categories: feature-based methods and model-based (end-to-end) learning methods [15].

1) *Feature-Based Detection*: Early methods focused on manually extracting low-level acoustic features from audio signals. Commonly used features include:

- Mel-Frequency Cepstral Coefficients (MFCC): Widely used in speech and speaker recognition, MFCC capture the timbral texture of an audio signal, making the useful for identifying unnatural spectral patterns in fake audio [16].
- Spectrograms: Visual representations of frequency over time are often analyzed using image-based models (e.g., CNN).
- Pitch, energy, jitter, shimmer: Variations in prosody and vocal stability can indicate synthesized speech.
- Temporal inconsistencies: Synthetic audio often includes unnatural pauses, abrupt transitions, or rhythm mismatches, which can be flagged as anomalies.

These features are typically fed into traditional machine learning classifiers such as Support Vector Machines (SVM), Random Forests, or shallow neural networks to distinguish between real and synthetic speech.

2) *Deep Learning-Based Detection*: More recent advances have leveraged deep learning models, particularly neural networks, to automatically learn relevant features from raw or preprocessed audio. These include:

- Convolution Neural Networks (CNN): Effective at analyzing spectrograms and capturing local spatial features that reveal synthesis artifacts [17].
- Recurrent Neural Networks (RNN) and their variants (e.g., LSTM, GRU): Designed to capture sequential dependencies and long-term temporal patterns in various audio representations [18].
- CNN + RNN Hybrid Models: Combining CNN for feature extraction and RNN for temporal modeling has proven effective in many studies.
- Transformer-based architectures: Models like the Audio Spectrogram Transformer (AST) and AVFakeNet [19] utilize attention mechanisms to focus on relevant portions of the input, improving performance on complex audio patterns.
- Multi-modal detection models: Some systems combine audio and visual cues to detect inconsistencies between voice and lip movement, though this is more applicable to video deepfakes [20].

C. Existing Dataset and its Challenges in Non-English Audio Detection

datasets are a crucial element for training and testing deepfake audio detectors. There are some benchmark datasets that have been widely utilized in the literature, such as ASVspoof, WaveFake, and FakeAVCeleb. These datasets feature synthetic and natural audio samples generated by various speech synthesis models such as Tacotron, WaveNet, and voice

conversion techniques. For example, the ASVspoof dataset series (2015, 2019, 2021) has been extremely popular for spoofing attack assessment on automatic speaker verification systems. WaveFake offers speech from GAN-based models, and FakeAVCeleb excels at multimodal deepfakes with audio and video components.

That is not to say, however, that there is no common weakness among these data sets. One of them is the English bias of speech and professional recording environments. The absence of acoustic and linguistic variation in training data limits a model’s ability to generalize to real-world, multilingual audio. Current systems struggle when applied to non-English languages or multi-accent scenarios [21].

This is particularly difficult in languages like Indonesian, where there is a single language (Bahasa Indonesia) with numerous traditional regional dialects, musical style, where real recordings will include background noise, code-switching, and pronunciation variation.

III. METHODOLOGY

A. Dataset

To build a model to identify AI-generated music in the Indonesian language context, we gathered a dataset of real and artificially generated audio. The real samples were taken from publicly available sources, primarily YouTube that covers a wide range of genres, artists, and recording qualities. This ensures the dataset represents real-world conditions in which AI-generated songs are typically shared.

The current state-of-the-art generative platforms enable us to easily collect artificial music. In particular, We explored Suno AI, which mostly used by individuals and communities specializing in generating songs across various genres. To complement this, we also collected AI-generated music from Youtube channels that focus on AI artist songs and explicitly claim to have used Suno AI for their generation. We ensured that our dataset was captured in variety of artificial composition, both directly from generative platforms and from real-world social media distributions

Additionally, to improve our generalization exposure, we created an additional cloned speech dataset in Bahasa Indonesia. This audio was generated using the latest voice cloning AI Voice Generator and Voice Cloning². In this dataset, the spoken words is exactly the same to the corresponding real recordings, but the voices are synthetically generated to imitate natural human speech. Even though this is not a music, the cloned speech serves as eval benchmark to assess how well our proposed detection models perform on unseen and different domain dataset within the same language highlighting the difficulties in this particular research area.

All the data that collected was in various length and uniform audio conditions. The diverse contributes to robust generalizable detection experiments for relatively unexplored field of studies and low-resources data language studies like Bahasa Indonesia.

²FineVoice by Fineshare: www.finevoice.fineshare.com

TABLE I
DATASET SUMMARY: NUMBER OF UNIQUE TITLES, CHUNKS, TOTAL DURATION, AND RECORDING QUALITY

Metric	AI (Fake)	Real
Number of Unique Titles	54	372
Number of Chunks	8413	9671
Sample Rate (Hz)	48000	44100
Channels	2	2
Bitrate (kbps)	1536	1411
Total Duration (s)	83876	96710

TABLE II
TOTAL DURATION OF SPEECH DATASET

Metric	AI (Fake)	Real
Number of Unique Titles	6	6
Number of Chunks	621	651
Sample Rate (Hz)	16000	16000
Channels	1	1
Bitrate (kbps)	256	256
Total Duration (s)	6210	6510

B. Data Preprocessing Techniques

The raw audio files that contain in our dataset have various duration, ranging from 20 minutes to over 3 hours. To maintain input length consistency for model training and batch processing, All audio files were segmented into fixed-length clips of 10 seconds duration. This duration was chosen to strike a practical balance: it is long enough to capture meaningful speech patterns and voice characteristics that are important to detect deepfake artifacts while having short enough duration to preserve manageable input size for deep learning pipelines which minimize GPU memory usage.

Each chunks is treated independent training sample, Allowing for scalable data better feature extraction. Instead of extracting time–frequency representations (e.g., Mel or log-Mel spectrograms), we use the fixed-length raw waveform itself as input, allowing the representations to be determined by the model itself. These representations then able to capture both long and short term acoustic range dependencies, Which are critical features to distinguish authentic from synthesized speech.

To avoid data leakage and speaker independent evaluation pipeline, the dataset is partitioned using an 80:10:10 split for training, validation, and testing, respectively. Crucially, no speaker appears in more than one subset; each recordings title are assigned exclusively to one of the three sets. This setup guarantees that model performance reflects true generalization to unseen voices, which is crucial requirement for reliable deepfake detection in real-world scenarios

All samples were collected from publicly available sources for research purposes only.

C. Classification Model

We make use of publicly available encoder as pretrained checkpoints as the backbone network. Specifically, we employ three state-of-the-art audio encoders: the Audio Spectrogram Transformer (AST) using the

MIT/ast-finetuned-audioset-10-10-0.4593 checkpoint, Wav2Vec 2.0 with the facebook/-wav2vec2-base checkpoint, and WavLM utilizing the microsoft/wavlm-base checkpoint. On top of the encoder, we attached a built in classification head from the Hugging Face, which automatically incorporate appropriate classification mechanism and include trainable linear projection layer that maps the encoder’s output representation into two classes: real and AI-generated. Both the encoder and the classification head is trained, where both parameters are updated using backpropagation [22], [23].

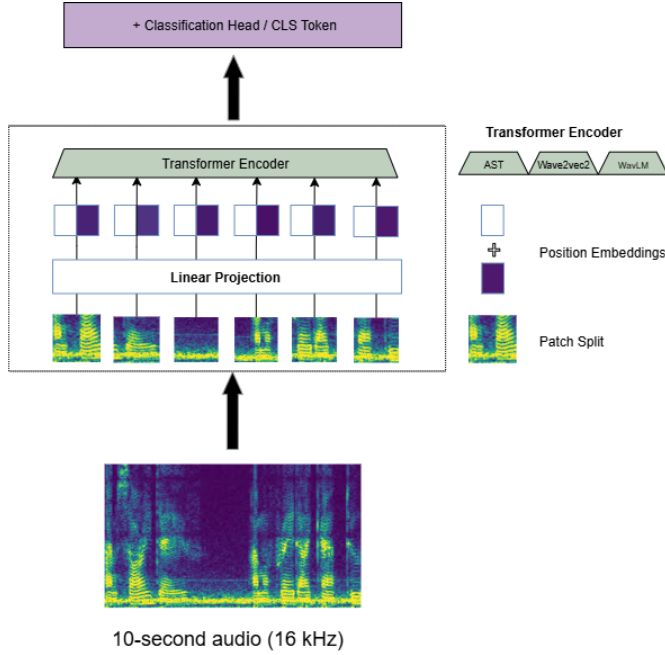


Fig. 1. Encoder-based audio classification pipeline

1) *AST*: The Audio Spectrogram Transformer (AST) is a transformer-based architecture adapted from vision Transformers (ViT) and was build for audio classification task using 2D time-frequency representations. Unlike waveform-based models such as Wave2Vec2 and WavLM, This model operates on spectrograms, treating them as visual inputs that capture both spectral and temporal patterns effectively.

This research, we focus on pretrained AST model that was fine-tuned for binary classification on our Indonesian deepfake audio dataset. Each of the 10-second audio chunks was converted into a log-Mel spectrogram. This resulting spectrograms are then normalized and reshaped to match AST input format. In Addition, a [CLS] token is added on top of the architecture which serves as the sequence level representation. A linear classification head is then connected to the [CLS] embedding to predict Real vs. Fake audio. .

AST uses self-attention mechanisms to capture long-range dependencies across time and frequency, making it suitable for detecting inconsistencies that was generated by deepfake synthesis. The training and evaluation strategy follows the same chunk-based pipeline that described earlier. The result

of the demonstrated AST later shows strong capacity to learn discriminative features from spectrograms, particularly under noisy and variable acoustic conditions [24].

2) *Wave2Vec2*: Wave2Vec 2.0 is a self-supervised learning model that was developed by Facebook. Unlike conventional approaches that need large amounts of labeled data, Wave2Vec2 making it particularly suitable for domains limited with annotated resources such as Indonesian speech dataset [25].

In this study, we implement the Wav2Vec2-ForSequenceClassification architecture, that adapts the pretrained Wave2Vec 2.0 encoder. This approach allows the model to directly learn from raw waveform features without rely on spectrogram transformations [26].

We utilize Hugging Face’s AutoProcessor to preprocess and normalize the waveform inputs. This step ensures consistent amplitude scaling and normalization, that prepares the data for transformer-based encoder. In this case, no separate text tokenizer was required since our task involves classification rather than transcription. The model makes predictions based on encoders output that are combined with mean pooling. Then, A linear classification head with two output neurons to predict is applied *Real* vs. *AI-generated*.

3) *Wavlm*: WavLM is a self-supervised speech representation model that was designed to extract and capture contextual rich features directly from raw audio waveforms. This model was build on HuBERT architecture, WavLM enhances performance by utilizing a larger training corpus and improved prediction strategies [27].

In this paper, we fine-tuned WavLM-Base+ variant for binary classification that was aimed to detect deepfake audio detection for Indonesian language. Unlike the conventional use of WavLM in automatic speech recognition (ASR), our study objective is to classify our dataset as either real or fake. With this purpose, we adopts the WavLMForSequenceClassification. This method adds a classification head on top of the pretrained WavLM encoder. Each of the audio files are resampled to 16 kHz and segmented into fixed-length 10 seconds audio chunks. This splitting method ensure various input size for training while maintaining enough acoustic content to capture deepfake-related features. If there is a segment that is shorter than 10 seconds for the end of the original audio, We drop the corresponding file to maintain consistent and original input length , instead of adding an empty placeholder.

We apply AutoFeatureExtractor to normalize and prepare the audio waveforms for the input to our model. Since the task does not involve transcription, no tokenizer is required. The model evaluated using accuracy, precision, recall, and F1-score. Our evaluation metrics is done at chunk level, with each 10-second segment treated as an independent input sample.

D. Fine-tuning approach

We trained each of our model with 3 epochs using the AdamW [28] optimizer with a learning rate of 3×10^{-5} ,

a batch size of 4 per device, and a weight decay of 0.01. Evaluation was performed on each epochs and steps while closely monitoring for overfitting signs in each steps.

In terms of hardware resources, all experiments were conducted on a consumer-grade RTX 4060 GPU with 8GB VRAM. Despite the limited computational resources, leveraging existing pretrained encoders enable us to perform a large-scale experiments on particularly limited resource and dataset, further demonstrating the practicality of our approach in limited-access environments.

E. Ethical and Legal Considerations

All our audio samples used in this study were collected from publicly accessible online sources and are used solely for non-commercial research purposes. Music Recordings obtained from YouTube are only included content that is publicly available under the platform’s terms of service. Raw audio instead will not be redistributed and any future release will be limited to metadata and source code to support reproducibility while respecting copyright and privacy expectations for online content creators. For cloned-speech dataset, all synthetic versions were generated only for controlled research experiments. Our goal is to support responsible use of AI in the Indonesian cultural and creative domain, models for AI-generated audio detection that can help platform and creators to identify misleading or unauthorized synthetic content. [29]

IV. EXPERIMENTS AND RESULTS

To evaluate the performance of the 3 different encoders on Indonesian Deepfake detection, we conducted experiments in two phases: (i) **in-domain evaluation** on the music dataset used for training, and (ii) **cross-domain evaluation** on a separate Indonesian speech dataset generated with the FineVoice synthesizer.

Two experimental settings were used:

- **Zero-shot:** we applied the original pretrained checkpoints from Hugging Face, replacing only the final classification layer with a randomly initialized 2-class head (real vs. fake) without any additional training on our dataset and tested on both of the dataset.
- **Fine-tuned:** We took the fine-tuned weights that was trained on our music dataset. Then, tested on both the in-domain (music) test set and the cross-domain (Cloned speech) dataset.

A. In-domain evaluation (Music dataset)

The primary training and evaluation were tested on the music dataset, which include both real and AI-generated samples. Each model was then trained on the training split and evaluated on the test split. The performance of each model was evaluate based on standard classification metrics: precision, recall, F1-score for both real and fake classes, and overall accuracy.

In the in-domain dataset (music), all models improve significantly after fine-tuning compared to the baseline zero-shot model. WavLM improved it’s accuracy from 59.4% to 96.3%,

TABLE III
CORE COMPARISON OF ZERO-SHOT AND FINE-TUNED EVALUATION ON SONG DATASET

Metric	WavLM		Wav2Vec 2.0		AST	
	Zero-Shot	Fine-Tuned	Zero-Shot	Fine-Tuned	Zero-Shot	Fine-Tuned
Precision (Real)	0.7017	0.9630	0.5737	0.9757	0.4789	0.8455
Precision (Fake)	0.5447	0.9624	0.6813	0.9618	0.2809	0.9969
Recall (Real)	0.4159	0.9672	0.9050	0.9662	0.6954	0.9978
Recall (Fake)	0.7980	0.9576	0.2319	0.9726	0.1359	0.7918
F1 (Real)	0.5223	0.9651	0.7022	0.9709	0.5672	0.9154
F1 (Fake)	0.6474	0.9600	0.3460	0.9671	0.1832	0.8826
Macro Avg (F1)	0.5849	0.9626	0.5241	0.9690	0.3752	0.8990
Accuracy	0.5943	0.9627	0.5908	0.9692	0.4342	0.9016

while Wave2Vec 2.0 increased from 59.1% to 96.9%. AST, although starts from lower zero-shot performance, reached 90.16% after fine-tuning. This results show that domain adaptation to Indonesia music improves each model capabilities on differentiate both real and AI-generated content.

B. Cross-domain evaluation (Speech dataset)

To assess the generalization capabilities of these models beyond the training dataset, we conducted experiments on a additional separate Indonesian speech dataset, which contains real and Cloned AI-generated speech audio. we performed the same experimental settings:

TABLE IV
COMPARISON OF ZERO-SHOT AND FINE-TUNED EVALUATION ON CROSS-DOMAIN (SPEECH) DATASET

Metric	WavLM		Wav2Vec 2.0		AST	
	Zero-Shot	Fine-Tuned	Zero-Shot	Fine-Tuned	Zero-Shot	Fine-Tuned
Precision (Real)	0.5068	0.5539	0.5515	0.6154	0.4193	0.6452
Precision (Fake)	0.0000	0.5831	0.5820	0.6524	0.0984	0.9003
Recall (Real)	0.9800	0.7496	0.7573	0.7373	0.6621	0.9524
Recall (Fake)	0.0000	0.3671	0.3543	0.5169	0.0386	0.4509
F1 (Real)	0.6681	0.6371	0.6382	0.6709	0.5134	0.7692
F1 (Fake)	0.0000	0.4506	0.4404	0.5768	0.0555	0.6009
Macro Avg (F1)	0.3340	0.5438	0.5393	0.6238	0.2844	0.6850
Accuracy	0.5016	0.5629	0.5605	0.6297	0.3577	0.7075

The key reason why the AST model shows better generalization results than other encoders in our cross-domain evaluation lie in its architecture and its pretraining strategy. AST is highly adapted from the Vision Transformer (ViT). It reuses almost the whole ViT, including the patch embedding, positional embeddings, and Transformer encoder blocks, from a ViT that has been pre-trained on the large-scale ImageNet dataset. This means that the cross-modal transfer from ImageNet substantially improves performance, especially with limited training data. The model inherits existing pattern-recognition capabilities on 2D structures that are adapted to time–frequency representations, namely spectrograms. As a result, once AST is fine-tuned on Indonesian music, it shows greater sensitivity to audio features such as harmonic structure, timbre, and texture that appear in both datasets, leading to stronger cross-domain generalization benchmark results compared to the other models. ³

³Code and extended experimental results:<https://github.com/JensonCHL/Conference>

V. CONCLUSION AND FUTURE WORK

This experiment reveals important insights on cross-domain generalization in low-resource language settings. Fine-tuning improves the overall detection performance, but the key finding relies on how each model performs across different domains that are still linguistically related. Results show that the pre-trained encoders, which were initially developed on English and general audio settings, struggle to be applied to Indonesian audio. After learning from Bahasa Indonesia characteristics, the performance across domains improves to a great extent, indicating that language-specific tuning plays a crucial role in achieving effective generalization. A further contribution of this work is the construction of a dedicated Indonesian dataset, providing more systematic benchmark for evaluating audio detection in the Indonesian context.

While results may be seen promising, several limitations appear in the process of undergoing this research. English and other major languages receive significant focus, but still Indonesian remains unexplored despite its large speaker base. This research gap is important, as the rise of AI-generated music in Indonesian context could have significant cultural, ethical, and security implications.

Future work should therefore focus on:

- Building larger and more diverse Indonesian real and AI-generated music datasets.
- Explicitly accounting for multi-accent variability in Bahasa Indonesia.
- Raising awareness of the research gap in Indonesian audio deepfake detection, thereby encouraging more attention from the global research community.

REFERENCES

- [1] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, and A. Défossez, "Simple and controllable music generation," in *Advances in Neural Information Processing Systems*, vol. 36. Curran Associates, Inc., 2023, neurIPS 2023. [Online]. Available: <https://github.com/facebookresearch/audiocraft>
- [2] P. M. Suhailudheen and K. M. Sheena, "Sun ai: Advancing ai-generated music with deep learning," *TechRxiv*, April 2025, preprint. [Online]. Available: <https://doi.org/10.36227/techrxiv.174417453.33743994/v1>
- [3] S. Prasetyo, Tongat, and N. Hidayah, "Identity theft and the rules in indonesia's criminal law," 01 2020.
- [4] I. G. A. G. A. Kadyanan, N. A. S. ER, A. A. I. N. E. Karyawati, I. G. N. A. W. Putra, M. S. Gunawan, N. M. J. Budiantari, and H. C. Octavia, "Balinese text-to-speech dataset as digital cultural heritage," *Data in Brief*, vol. 60, p. 111528, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352340925002604>
- [5] I. J. Gunawan and S. Janisriwati, "Legal analysis on the use of deepfake technology: threats to indonesia banking institutions," *Law and Justice*, vol. 8, no. 2, pp. 192–210, 2023.
- [6] A. Purwadi, C. Y. Serfiyani, and C. R. Serfiyani, "Legal landscape on national cybersecurity capacity in combating cyberterrorism using deep fake technology in indonesia," *International Journal of Cyber Criminology*, vol. 16, no. 1, 2022.
- [7] G. Sariasa, N. L. M. Madhusodani, D. A. D. A. A.P, and N. P. E. Parwati, "The existence of deepfake artificial intelligence as a means of crime in the perspective of indonesia's positive law," *EAI*, 12 2024.
- [8] H. Dinkel, Z. Yan, Y. Wang, J. Zhang, Y. Wang, and B. Wang, "Scaling up masked audio encoder learning for general audio classification," *arXiv preprint arXiv:2406.06992*, 2024.
- [9] F. Gontier, R. Serizel, and C. Cerisara, "Automated audio captioning by fine-tuning bart with audioset tags," in *DCASE 2021-6th Workshop on Detection and Classification of Acoustic Scenes and Events*, 2021.
- [10] R. Arvin, A. M. T. Anggraini, and W. Retnowulandari, "Regulatory challenges of digital direct licensing on royalty in indonesia's copyright regime," *JL Pol'y & Globalization*, vol. 148, p. 46, 2025.
- [11] M. N. Febrian, R. Setyawati, A. Noventri, and Y. Mahendra, "Review of legal certainty of royalty collection on music and/or songs: A comparative study of indonesia and russia," *Realism: Law Review*, vol. 3, no. 1, pp. 115–143, 2025.
- [12] L.-C. Yang, S.-Y. Chou, and Y.-H. Yang, "Midinet: A convolutional generative adversarial network for symbolic-domain music generation," in *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*. Suzhou, China: ISMIR, 2017, arXiv:1703.10847. [Online]. Available: <https://arxiv.org/abs/1703.10847>
- [13] A. Muhamed, L. Li, X. Shi, S. Yaddanapudi, W. Chi, D. Jackson, R. Suresh, Z. C. Lipton, and A. J. Smola, "Symbolic music generation with transformer-gans," in *Proceedings of the Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21)*. AAAI, 2021, pp. 408–417.
- [14] S. Mangal, R. Modak, and P. Joshi, "Lstm based music generation system," *IARJSET*, vol. 6, no. 5, p. 47–54, May 2019. [Online]. Available: <http://dx.doi.org/10.17148/IARJSET.2019.6508>
- [15] R. Singh, O. Das, P. Jain, and R. Sinha, "Development of a multimodal framework for deepfake detection," in *Proceedings of the 10th International Conference on Machine Vision and Machine Learning (MVML 2024)*. Avestia Publishing, 2024, p. 115. [Online]. Available: <https://doi.org/10.11159/mvml24.115>
- [16] A. Hamza, A. R. R. Javed, F. Iqbal, N. Kryvinska, A. S. Almadhor, Z. Jalil, and R. Borghol, "Deepfake audio detection via mfcc features using machine learning," *IEEE Access*, vol. 10, pp. 134 018–134 028, 2022.
- [17] J. Khochare, C. Joshi, B. Yenarkar, S. Suratkar, and F. Kazi, "A deep learning framework for audio deepfake detection," *Arabian Journal for Science and Engineering*, vol. 47, no. 3, pp. 3447–3458, 2022.
- [18] A. Chintha, B. Thai, S. J. Sohrawardi, K. Bhatt, A. Hickerson, M. Wright, and R. Ptucha, "Recurrent convolutional structures for audio spoof and video deepfake detection," *IEEE Journal of Selected Topics in Signal Processing*, vol. 14, no. 5, pp. 1024–1037, 2020.
- [19] H. Ilyas, A. Javed, and K. M. Malik, "Avfakenet: A unified end-to-end dense swin transformer deep learning model for audio–visual deepfakes detection," *Applied Soft Computing*, vol. 136, p. 110124, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1568494623001424>
- [20] Y. Kim and S. Go, "Segment transformer: Ai-generated music detection via music structural analysis," 2025. [Online]. Available: <https://arxiv.org/abs/2509.08283>
- [21] Z. Almutairi and H. Elgibreen, "A review of modern audio deepfake detection methods: challenges and future directions," *Algorithms*, vol. 15, no. 5, p. 155, 2022.
- [22] H. Yang, J. Zhao, G. Haffari, and E. Shareghi, "Investigating pre-trained audio encoders in the low-resource condition," *arXiv preprint arXiv:2305.17733*, 2023.
- [23] B. Kim, A. Bydlon, Q. Tang, H. Phan, C.-C. Kao, T. Zhang, and C. Wang, "Effective techniques for scaling audio encoder pretraining," in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [24] Y. Gong, Y.-A. Chung, and J. Glass, "Ast: Audio spectrogram transformer," 4 2021.
- [25] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," 6 2020.
- [26] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "Hubert: Self-supervised speech representation learning by masked prediction of hidden units," 6 2021.
- [27] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, X. Yu, and F. Wei, "Wavlm: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, p. 1505–1518, Oct. 2022. [Online]. Available: <http://dx.doi.org/10.1109/JSTSP.2022.3188113>
- [28] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2019. [Online]. Available: <https://arxiv.org/abs/1711.05101>
- [29] W. Sui, A. Sui, and R. E. Rhodes, "What to watch: Practical considerations and strategies for using youtube for research," *DIGITAL HEALTH*, vol. 8, p. 20552076221123707, 2022. [Online]. Available: <https://doi.org/10.1177/20552076221123707>